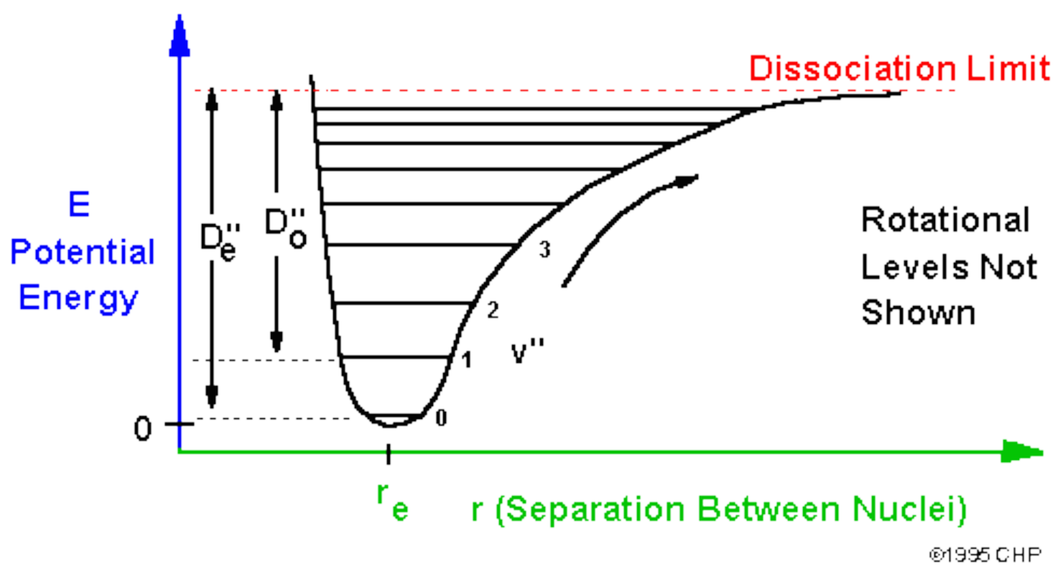
The Diatomic Molecule

John W Daily

The Born-Oppenheimer Approximation and the Diatomic Molecule

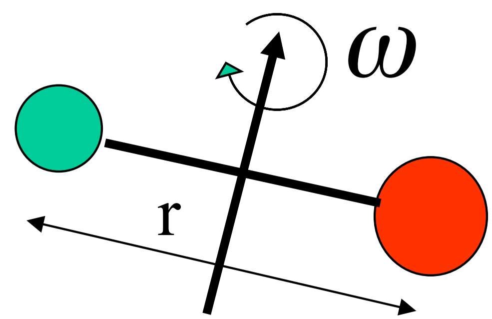
Electrons move fast, nuclei move slowly (think about Pigpen)



We can pretend that there is a potential force field that gives rise to an effective potential

Rigid Rotator and Harmonic Oscillator Approximations

- Rigid Rotator
  - Here we assume the molecule is rigidly bonded.

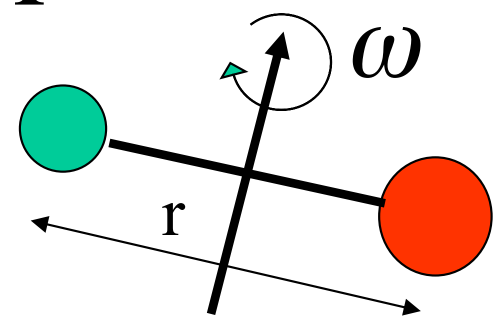


- Harmonic Oscillator
  - Here we assume the bond acts like a simple linear spring $F = k(r - r_e)$
  - Hence

$$V(r) = \frac{k(r - r_e)^2}{2}$$

where r_e is the equilibrium value of the radius

Rigid Rotator



The angular momentum is

$$L = I\omega$$

where I is the moment of inertia $I = \mu r^2$

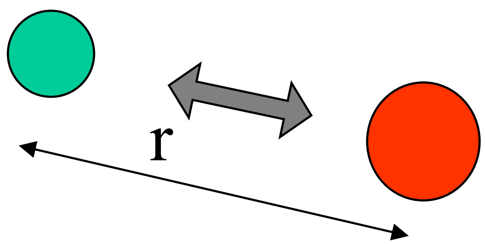
Then the energy is $\varepsilon_R = \frac{1}{2} I\omega^2 = \frac{L^2}{2I} = l(l+1)\frac{\hbar^2}{2\mu r^2}$

For molecular rotation it is common to use J as the rotational quantum number, so

$$\varepsilon_R = J(J+1)\frac{\hbar^2}{2\mu r^2}$$

Harmonic Oscillator

The radial wave equation becomes



$$\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left\{\varepsilon-V(r)-\frac{\hbar^2}{2\mu r^2}l(l+1)\right\}R=0$$

where $$V(r)=\frac{k(r-r_e)^2}{2}$$

Harmonic Oscillator

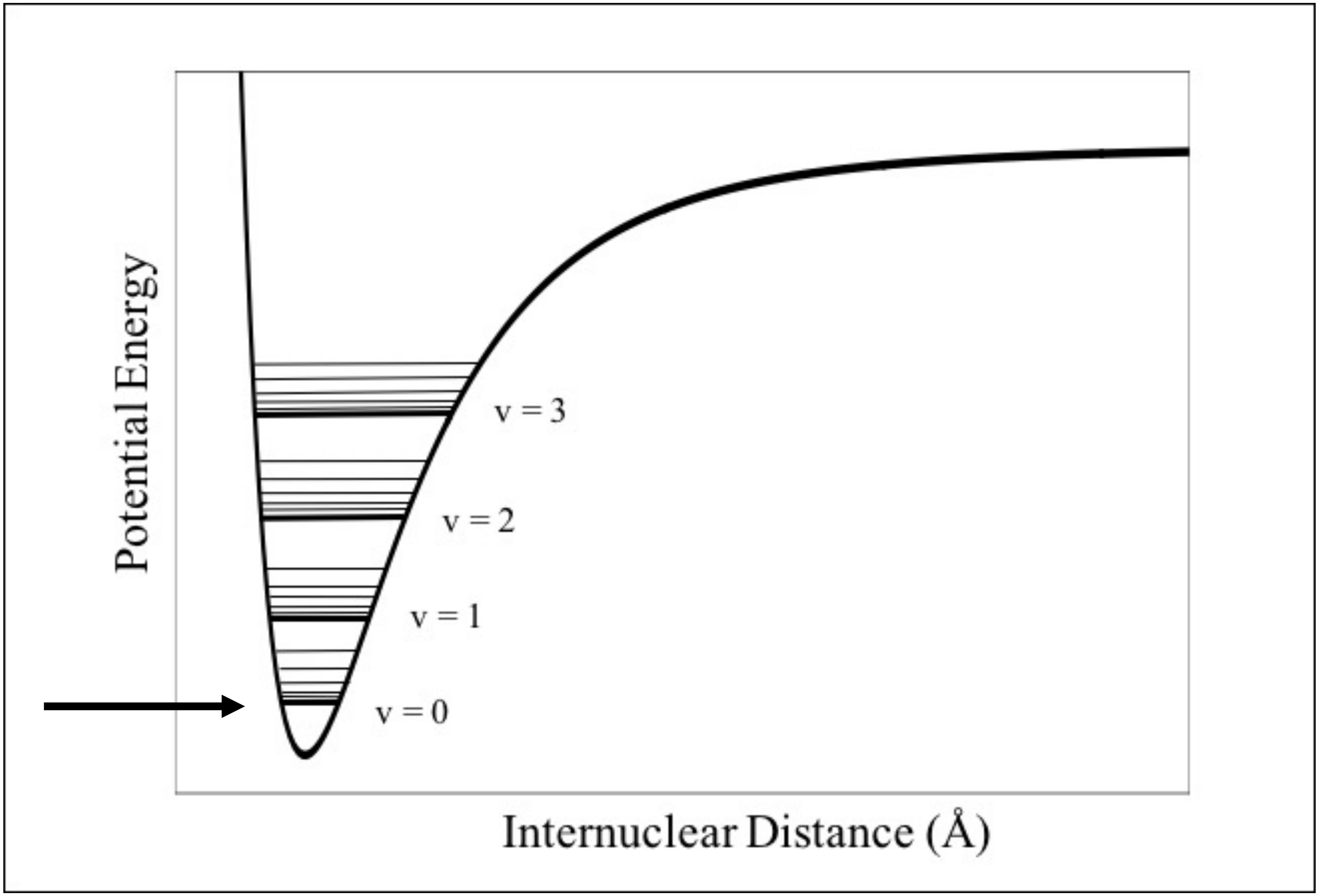
- This is a Hermite equation and the solution for the energy is

$$\varepsilon_v = (\mathrm{v} + \frac{1}{2})h\nu_v \qquad \mathrm{v} = 0,1,2,3,...$$

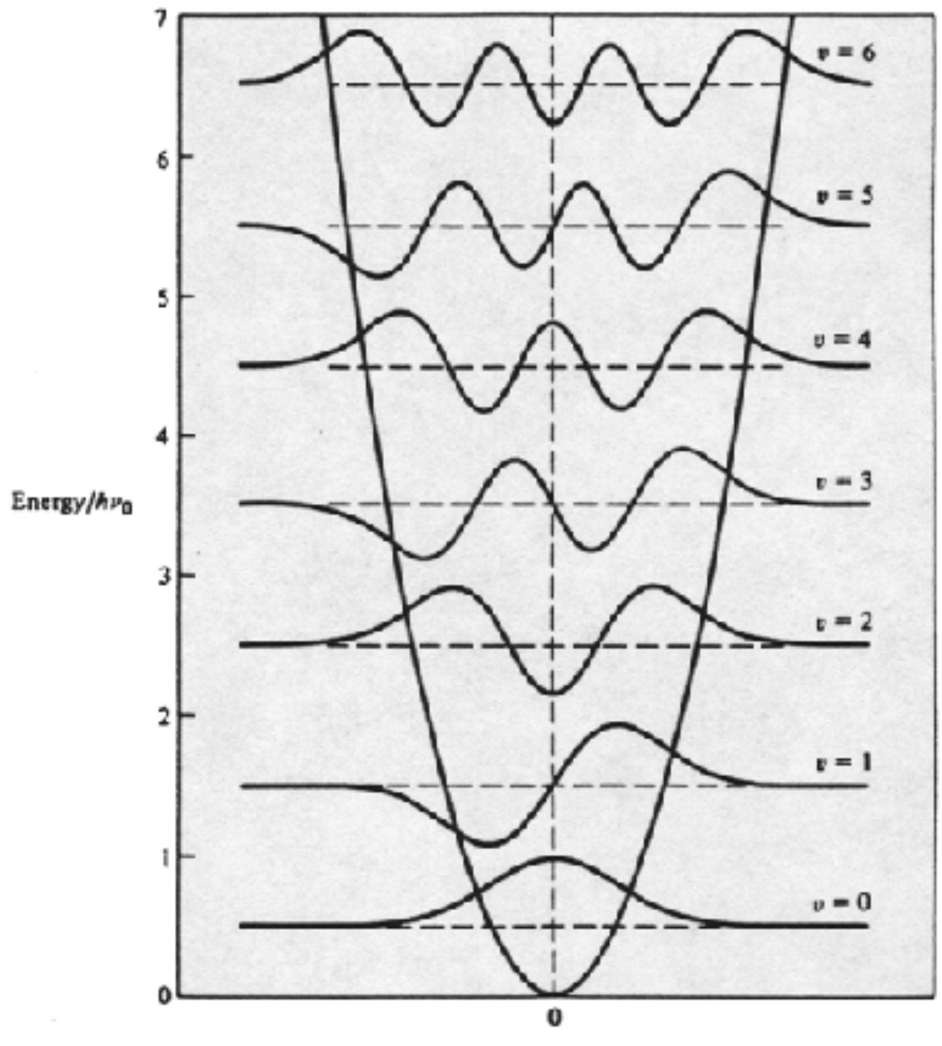
- Note that energy does NOT go to zero. This is called the zero point energy.
- The degeneracy is unity.

Result

ZPE



Wave Functions



The Diatomic Molecule

The total rotational and vibrational energy is then

$$\varepsilon_R + \varepsilon_v = \frac{\hbar^2}{2I_e} J(J+1) + h\nu_v(\mathrm{v} + \frac{1}{2})$$

J is the same as l and $J = 0,1,2,3,...$ and $\mathrm{v} = 0,1,2,3,...$

The degeneracies of these energies are:

$$g_J = 2J + 1$$

$$g_v = 1$$

The Diatomic Molecule

The quantum state energies are more commonly expressed as

$$F(J) = \frac{\varepsilon_R}{hc} = B_e J(J+1) \qquad G(v) = \frac{\varepsilon_v}{hc} = \omega_e(v + \frac{1}{2})$$

Where the units are cm^{-1}. This nomenclature was developed by spectroscopists who noted that the inverse of the wavelength of spectroscopic transitions is proportional to the energy difference between the two quantum states involved in the transition because:

$$\varepsilon = h\nu = h\frac{c}{\lambda}$$

The Diatomic Molecule

Alternatively, we can express the energy in terms of characteristic temperatures

$$T_R \equiv \frac{\hbar^2}{2I_e k} = \frac{B_e}{(k/hc)} \qquad T_V \equiv \frac{h\nu_v}{k} = \frac{\omega_e}{(k/hc)}$$

For N_2 $\quad B_e = 1.998\ \text{cm}^{-1} \qquad T_R = 2.87\ \text{K}$

$$\omega_e = 2357.6\ \text{cm}^{-1} \quad T_V = 3390\ \text{K}$$

So, $B_e << \omega_e$

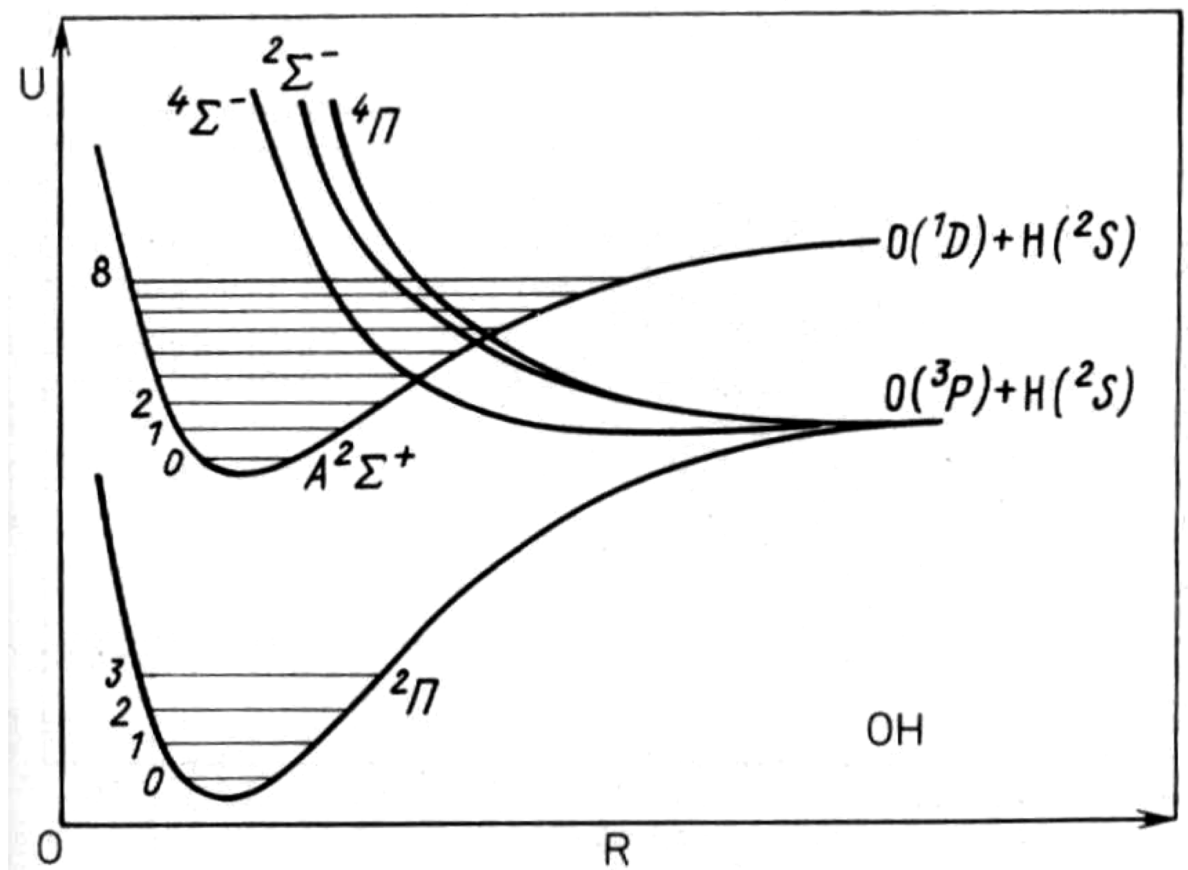
Real Diatomic Molecules

In real molecules, the rigid rotator/harmonic oscillator approximation is not exactly valid. Often one can correct for deviations from the simple theory using polynomial expansions in terms of the quantum numbers:

$$G(v) + F(J) = \omega_e(v + \frac{1}{2}) - \omega_e x_e(v + \frac{1}{2})^2 + \omega_e y_e(v + \frac{1}{2})^3 - \ldots$$
$$+ B_v J(J+1) - D_v J^2(J+1)^2 + \ldots$$

The higher order coefficients are typically much smaller than the first order coefficients.

Real Diatomic Molecules



Typical Diatomic Constants

Ground Electronic State Data

Table 7.1 Molecular Constants for Selected Diatomic Molecules in the Electronic Ground State (From JANAF Thermochemical Tables NSRDS-NBS 37)

Molecule	$B_e(\mathrm{cm}^{-1})$	$\alpha_e(\mathrm{cm}^{-1})$	$\omega_e(\mathrm{cm}^{-1})$	$\omega_e x_e(\mathrm{cm}^{-1})$	$T_r(^\circ\mathrm{K})$	$T_v(^\circ\mathrm{K})$	$r_e(10^{-8}\ \mathrm{cm})$	$D_o(\mathrm{ev})$
Cl_2	0.2408	0.0017	561.1	4.00	0.35	807	1.986	2.47
CO	1.9302	0.0175	2169.5	13.45	2.78	3120	1.128	11.10
H_2	60.848	3.066	4405.3	125.33	80.5	6345	0.742	4.48
HCl	10.588	0.3037	2987.6	52.06	15.2	4300	1.275	4.43
HF	20.956	0.798	4138.3	89.88	30.2	5959	0.917	5.84
HI	6.512	0.172	2309.1	39.73	9.37	3320	1.604	3.06
I_2	0.037	0.0001	214.5	0.61	0.05	309	2.666	1.54
N_2	1.998	0.0179	2357.6	14.06	2.87	3390	1.088	9.76
NO	1.704	0.0178	1903.6	13.97	2.45	2740	1.151	6.48
O_2	1.445	0.0158	1580.2	12.07	2.08	2280	1.207	5.12

http://webbook.nist.gov/chemistry/

Units

$$1 \text{ e.v.} = 8058 \text{ cm}^{-1} = 11,600 \text{ K} = 1.602\text{e-}19 \text{ J} = 23.06 \text{ kcal/mol}$$